THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 5: **Prime Time** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 10	$48 = 2^4 \times 3$; use $(e_1+1)(e_2+1)\ldots$
2	**(a)** 12	Highest common factor.
3	**(c)** 36	Lowest common multiple.
4	**(c)** 1	Digit sum must be a multiple of 3.
5	**(b)** 994	Divide 999 by the number.
6	**(a)** 9	Count them carefully.
7	**(a)** 17	$210 = 2 \times 3 \times 5 \times 7$
8	**(c)** 8	$54 = 2 \times 3^3$; use $(e_1+1)(e_2+1)\ldots$
9	**(c)** 15	Highest common factor.
10	**(a)** 40	Lowest common multiple.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$24 = 2 \times 2 \times 2 \times 3$, $36 = 2 \times 2 \times 3 \times 3$ (1); HCF $= 12$, LCM $= 72$ (1).	1+1
12	1234 divided by 9 leaves remainder 1 (1); Add 8 to get 1242 (1).	1+1
13	Digit sum of known digits is 15 (1); Missing digit $= 3$ (1).	1+1
14	$30 = 2 \times 3 \times 5$, $45 = 3 \times 3 \times 5$ (1); HCF $= 15$, LCM $= 90$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$180 = 2^2 \times 3^2 \times 5$ (1); Number of factors $= (2+1) \times (2+1) \times (1+1) = 18$ (2).	1+2
16	LCM of 12 and 18 is 36 (1); After 36 minutes (1); Time: 9:36 (1).	1+1+1
17	Length of piece $= \mathrm{HCF}(48, 72) = 24$ m (1); Pieces from the first rope: 2; from the second: 3 (1); Total pieces $= 5$ (1).	1+1+1
	OR: $200 = 2^3 \times 5^2$ (1); Number of factors $= (3+1) \times (2+1) = 12$ (2).	1+2
18	LCM of 15 and 20 is 60 (1); After 60 minutes (1); Time: 10:00 (1).	1+1+1
	OR: Length of piece $= \mathrm{HCF}(60, 84) = 12$ m (1); Pieces from the first rope: 5; from the second: 7 (1); Total pieces $= 12$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Side $= \mathrm{HCF}(450, 360) = 90$ cm (1); Along the length: 5 tiles (1); Along the width: 4 tiles (1); Total $= 20$ tiles (2).	1+1+1+2
	OR: LCM of 4, 5, 6 $= 60$ (1); Smallest number $= 61$ (1); Next number $= 121$ (1); Check remainders (2).	1+1+1+2
20	Side $= \mathrm{HCF}(540, 420) = 60$ cm (1); Along the length: 9 tiles (1); Along the width: 7 tiles (1); Total $= 63$ tiles (2).	1+1+1+2
	OR: LCM of 3, 4, 8 $= 24$ (1); Smallest number $= 25$ (1); Next number $= 49$ (1); Check remainders (2).	1+1+1+2

Mistakes to avoid:

- Calling 1 a prime number; 1 is neither prime nor composite.
- Thinking all odd numbers are prime (9, 15, 21, 91 are composite) or that 2 is not prime.
- Taking the greatest powers for HCF or the smallest powers for LCM; it is the other way round.
- Using LCM when the question asks for the largest equal size (HCF), and the reverse for 'meet again'.
- Applying the divisibility test for 4 or 8 to the digit sum instead of the last two or three digits.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 5: **Prime Time** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 2	Digit sum must be a multiple of 3.
2	**(c)** 988	Divide 999 by the number.
3	**(c)** 9	Count them carefully.
4	**(b)** 21	$330 = 2 \times 3 \times 5 \times 11$
5	**(d)** 12	$72 = 2^3 \times 3^2$; use $(e_1 + 1)(e_2 + 1) \ldots$
6	**(c)** 28	Highest common factor.
7	**(c)** 75	Lowest common multiple.
8	**(d)** 0	Digit sum must be a multiple of 3.
9	**(d)** 986	Divide 999 by the number.
10	**(a)** 12	Count them carefully.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	5678 divided by 11 leaves remainder 2 (1); Add 9 to get 5687 (1).	1+1
12	Digit sum of known digits is 19 (1); Missing digit $= 8$ (1).	1+1
13	$42 = 2 \times 3 \times 7$, $56 = 2 \times 2 \times 2 \times 7$ (1); HCF $= 14$, LCM $= 168$ (1).	1+1
14	3456 divided by 7 leaves remainder 5 (1); Add 2 to get 3458 (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$252 = 2^2 \times 3^2 \times 7$ (1); Number of factors $= (2+1) \times (2+1) \times (1+1) = 18$ (2).	1+2
16	LCM of 20 and 30 is 60 (1); After 60 minutes (1); Time: 10:00 (1).	1+1+1
17	Length of piece $= \text{HCF}(90, 126) = 18$ m (1); Pieces from the first rope: 5; from the second: 7 (1); Total pieces $= 12$ (1).	1+1+1
	OR: $300 = 2^2 \times 3 \times 5^2$ (1); Number of factors $= (2+1) \times (1+1) \times (2+1) = 18$ (2).	1+2
18	LCM of 24 and 36 is 72 (1); After 72 minutes (1); Time: 10:12 (1).	1+1+1
	OR: Length of piece $= \text{HCF}(56, 98) = 14$ m (1); Pieces from the first rope: 4; from the second: 7 (1); Total pieces $= 11$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Side $= \text{HCF}(600, 480) = 120$ cm (1); Along the length: 5 tiles (1); Along the width: 4 tiles (1); Total $= 20$ tiles (2).	1+1+1+2
	OR: LCM of 5, 6, 10 $= 30$ (1); Smallest number $= 31$ (1); Next number $= 61$ (1); Check remainders (2).	1+1+1+2
20	Side $= \text{HCF}(390, 330) = 30$ cm (1); Along the length: 13 tiles (1); Along the width: 11 tiles (1); Total $= 143$ tiles (2).	1+1+1+2
	OR: LCM of 6, 8, 9 $= 72$ (1); Smallest number $= 73$ (1); Next number $= 145$ (1); Check remainders (2).	1+1+1+2

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- Taking the greatest powers for HCF or the smallest powers for LCM; it is the other way round.
- Using LCM when the question asks for the largest equal size (HCF), and the reverse for 'meet again'.
- Applying the divisibility test for 4 or 8 to the digit sum instead of the last two or three digits.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 3

Chapter 5: **Prime Time** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 23	$390 = 2 \times 3 \times 5 \times 13$
2	**(b)** 9	$100 = 2^2 \times 5^2$; use $(e_1+1)(e_2+1)\ldots$
3	**(a)** 7	Highest common factor.
4	**(c)** 70	Lowest common multiple.
5	**(b)** 2	Digit sum must be a multiple of 3.
6	**(d)** 989	Divide 999 by the number.
7	**(a)** 10	Count them carefully.
8	**(d)** 23	$462 = 2 \times 3 \times 7 \times 11$
9	**(a)** 16	$120 = 2^3 \times 3 \times 5$; use $(e_1+1)(e_2+1)\ldots$
10	**(d)** 18	Highest common factor.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Digit sum of known digits is 16 (1); Missing digit $= 2$ (1).	1+1
12	$48 = 2 \times 2 \times 2 \times 2 \times 3$, $72 = 2 \times 2 \times 2 \times 3 \times 3$ (1); HCF $= 24$, LCM $= 144$ (1).	1+1
13	7891 divided by 8 leaves remainder 3 (1); Add 5 to get 7896 (1).	1+1
14	Digit sum of known digits is 13 (1); Missing digit $= 5$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$392 = 2^3 \times 7^2$ (1); Number of factors $= (3+1) \times (2+1) = 12$ (2).	1+2
16	LCM of 10 and 25 is 50 (1); After 50 minutes (1); Time: 9:50 (1).	1+1+1
17	Length of piece $= \mathrm{HCF}(75, 100) = 25$ m (1); Pieces from the first rope: 3; from the second: 4 (1); Total pieces $= 7$ (1).	1+1+1
	OR: $450 = 2 \times 3^2 \times 5^2$ (1); Number of factors $= (1+1) \times (2+1) \times (2+1) = 18$ (2).	1+2
18	LCM of 16 and 20 is 80 (1); After 80 minutes (1); Time: 10:20 (1).	1+1+1
	OR: Length of piece $= \mathrm{HCF}(105, 135) = 15$ m (1); Pieces from the first rope: 7; from the second: 9 (1); Total pieces $= 16$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Side $= \mathrm{HCF}(720, 540) = 180$ cm (1); Along the length: 4 tiles (1); Along the width: 3 tiles (1); Total $= 12$ tiles (2).	1+1+1+2
	OR: LCM of 4, 6, 9 $= 36$ (1); Smallest number $= 37$ (1); Next number $= 73$ (1); Check remainders (2).	1+1+1+2
20	Side $= \mathrm{HCF}(840, 630) = 210$ cm (1); Along the length: 4 tiles (1); Along the width: 3 tiles (1); Total $= 12$ tiles (2).	1+1+1+2
	OR: LCM of 3, 5, 9 $= 45$ (1); Smallest number $= 46$ (1); Next number $= 91$ (1); Check remainders (2).	1+1+1+2

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